ARROWSTACK FULL STACK DEVELOPMENT INTERNSHIP

Project 1 — B2B Order Management Platform

Type: Foundation / Core industry task **Stack:** Node.js + Express REST API · JWT + RBAC · relational store with SQL migrations · React (Vite) SPA

Scenario

A simulated distributor sells products to business buyers through a multi-role ordering workflow: catalogue -> cart -> checkout -> order -> invoice, with stock control and admin oversight.

Roles

Role	Can do
buyer	browse catalogue, manage cart, place orders, view own orders & invoices
seller	create/update products, view all orders, move orders through fulfilment, view stats
admin	everything above plus manage users (activate/deactivate) and read the audit log

Core features (per the handbook)

- **Role-based access** — enforced server-side with `requireRole`.
- **REST API integration** — JSON REST endpoints consumed by the React SPA.
- **Relational database with migrations** — five versioned migrations (users, products, carts, orders, invoices).
- **Validation & error handling** — declarative schemas; 400/401/403/404/409 responses with clear messages.
- **Audit-friendly activity records** — every mutation written to `audit_log`.
- **Deployment-ready configuration** — `PORT`, `DB_FILE`, `JWT_SECRET` via environment variables.

Demo logins

Email	Password	Role
admin@arrowstack.in	Admin@123	admin
seller@arrowstack.in	Seller@123	seller
buyer@arrowstack.in	Buyer@123	buyer

Run it

```
# backend (from repo root)
npm install
npm run start:p1            # http://localhost:4001

# frontend (new terminal)
cd project-1-b2b-order-management/web
npm install
npm run dev                 # http://localhost:5171  (proxies /api -> 4001)
```

API summary

Method	Path	Access	Purpose
POST	`/api/auth/register`	public	register a buyer or seller
POST	`/api/auth/login`	public	obtain JWT
GET	`/api/auth/me`	any	current user
GET	`/api/products`	any	list active products
POST/PATCH	`/api/products` `/api/products/:id`	seller, admin	manage catalogue
GET	`/api/cart`	buyer	view cart
POST	`/api/cart/items`	buyer	add item (stock-checked)
DELETE	`/api/cart/items/:id`	buyer	remove item
POST	`/api/orders`	buyer	checkout cart -> order + invoice
GET	`/api/orders` `/api/orders/:id`	any (scoped)	view orders
PATCH	`/api/orders/:id/status`	seller, admin	fulfil / cancel
GET	`/api/invoices`	any (scoped)	list invoices
GET/PATCH	`/api/admin/users` `/api/admin/users/:id`	admin	user management
GET	`/api/admin/audit`	admin	audit trail
GET	`/api/admin/stats`	seller, admin	order & stock stats

Business rules

- GST applied at **18%** on the order subtotal; total = subtotal + GST.
- Adding to cart checks available stock; over-quantity adds are rejected (409).
- Checkout decrements stock atomically per line item and issues an invoice (`INV-<year>-<id>`).
- Buyers only ever see their own orders/invoices.

Documentation

- Requirements (docs/REQUIREMENTS.md)
- Test evidence (docs/TEST-EVIDENCE.md) — 23/23 checks passing

Source: `server/` (API), `web/` (React SPA), `test/run.js` (validation scenario).